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MILITARY SPECIFICATION

METERS, FLOW, MECHANICAL, VOLUMETRIC POSITIVE DISPLACEMENT, LIQUID, COLD WATER AND HOT WATER TYPE, NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fresh water mechanical flowmeters (cold and hot water service) of the positive displacement type.

1.2 Classification. Flowmeters shall be of the following classes, as specified (see 6.2.1):

Class A - Volumetric flow measurement of cold fresh water up to 100 degrees Fahrenheit (°F).
Class B - Volumetric flow measurement of hot fresh water from 100 to 180°F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6680

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SPECIFICATIONS

FEDERAL

A-A-1683 - Tape, Pressure-Sensitive Adhesive (Packaging, Paper).
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/2 - Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes, and General Application.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.24 - Bronze Pipe Flanges and Flanged Fittings, Class 150 and 300.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The flowmeter shall consist of a volumetric-type measuring drum, or a fixed displacement device, and a mechanically or magnetically actuated registering element. The assembly comprising the flowmeter shall be equipped for installation in a specified piping system. The liquid shall be measured by isolating the flow either in a predetermined volume space or in a fixed displacement mechanism, such as a piston assembly or nutating disk assembly. Flowmeter parts in contact with the measured liquid shall be composed of materials that will not chemically react with the liquid. Unless otherwise specified (see 6.2.1), the flowmeter shall not be supplied with a resettable counter. The flowmeter parts shall be arranged to operate within the accuracy requirements specified herein, and shall be readily accessible for maintenance and replacement without disturbing the connections to the pipe lines.

3.2 Approved standard commercial product. The flowmeter shall be, as a minimum, the manufacturer's approved standard commercial product (see 6.5.1) and shall meet the requirements of this specification. Additional or improved features not specifically prohibited by this specification but which are a part of the manufacturer's approved standard commercial product may be included.

3.2.1 Repair parts and service. Replacement or repair parts and service for the approved standard commercial product provided under this specification shall be available from the manufacturer, through commercial parts distribution, or service organizations.

3.3 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.4 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality normally used in the manufacturer's approved standard practice. Material shall not have been in prior service except as required by manufacturer's tests.

3.4.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. When such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fasteners shall be of the same material as the metals joined, or shall be cathodic to the metals joined.

3.4.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.5 Construction. The flowmeters shall be constructed to maintain accuracy of indication and to resist wear and corrosion. Materials for construction of the pressure-containing parts shall be selected to be compatible with the pressure and temperature ratings of this specification. Moving parts shall be properly aligned, balanced, and adequately lubricated. Interior parts, seals, and packings shall be accessible for service without disturbing the connections to the pipe lines.

3.6 Components. The flowmeter shall consist of additional apparatus normally furnished with this type of flowmeter including the components specified in 3.6.1 through 3.6.8.

3.6.1 Flowmeter case. The flowmeter case shall be constructed of a non-ferrous, non-magnetic, and non-corrosive alloy.

3.6.2 Register case. The register case and lid shall be constructed of a non-ferrous, non-magnetic, and non-corrosive alloy. In addition, it shall be removable but securely fastened to the flowmeter case so as not to present a missile hazard under shock conditions specified in 4.6.4 and to ensure maximum protection to the register. The register window shall be fabricated from a clear, non-fogging material of sufficient size to allow for visualization of the entire register face.

3.6.3 Measuring chamber. The measuring chamber shall be designed to maintain a constant volume displacement or constant volumetric measurement with each cycle of the meter. Unless otherwise specified (see 6.2.1), it shall be constructed of the manufacturer's approved standard material.

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3.6.4 Piston or disk measuring elements. Piston or disk-type measuring elements shall be supported and aligned to prevent twisting, bending, and distortion. Piston nutations or oscillations shall be kept to a minimum, consistent with standard industry practice.

3.6.5 Register. Unless otherwise specified (see 6.2.1), the register shall be of the horizontal type, with figures reading in United States liquid gallons direct. The respective maximum indication unit of the initial register circle or wheel, and the respective minimum indication of the register, shall be as specified in table I. The flowmeter designations specified in table I are representative of ranges of volumetric flow rates for class A and class B flowmeters as specified in 3.7.1.2 and 3.7.2.2, respectively. The flowmeter designations do not necessarily correspond to a manufacturer's flowmeter designation which is specified in terms of line, pipe, or model size. The correspondence between flowmeter designation and nominal pipe size is specified in 3.6.8.

TABLE I. Flowmeter register indication.

Flowmeter designation	Maximum indication unit of initial register circle or wheel (gallons)	Minimum indication of register (gallons)
1	10	999,990
2	10	999,990
3	10	999,990
4	100	999,900
5	100	9,999,900
6	100	9,999,900

3.6.6 Strainer. Unless otherwise specified (see 6.2.1), the flowmeter shall be furnished with integral strainers.

3.6.7 Check valve. When a reversal of flow through the flowmeter might result in damage, the flowmeter shall be furnished with a check valve or other means for preventing such flow.

3.6.8 Piping connections. Unless otherwise specified (see 6.2.1), class A and class B flowmeters shall have end connections in accordance with table II. Screw threads shall be in accordance with FED-STD-H28 and FED-STD-H28/2. Flanged connections shall be in accordance with ANSI B16.24 for 150 pound bronze flanges. Flange connections shall be machined on faces and rim, and bolt holes spotfaced and drilled in accordance with table II.

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TABLE II. Flowmeter end connections.

Flowmeter designation	Nominal pipe size (inches)	End connections	Flange characteristics				
			Outer diameter (inches)	Thickness (inches)	Bolt circle (inches)	Bolt hole diameter (inches)	Number of bolt holes
1	3/4	NPT	-	-	-	-	-
2	1	NPT	-	-	-	-	-
3	1-1/4	NPT or Flange	4-5/8	13/32	3-1/2	5/8	4
3	1-1/2	NPT or Flange	5	7/16	3-7/8	5/8	4
3 or 4	2	NPT or Flange	6	1/2	4-3/4	3/4	4
4	2-1/2	Flange	7	9/16	5-1/2	3/4	4
4	3	Flange	7-1/2	5/8	6	3/4	4
4	3-1/2	Flange	8-1/2	11/16	7	3/4	8
5	4	Flange	9	11/16	7-1/2	3/4	8
6	6	Flange	11	13/16	9-1/2	7/8	8

3.7 Performance characteristics.

3.7.1 Class A flowmeters.

3.7.1.1 Working pressure. Unless otherwise specified (see 6.2.1), class A flowmeters shall be capable of a minimum working pressure of 150 pounds per square inch (lb/in^2) (see 4.6.1).

3.7.1.2 Registration. The registration of class A flowmeters shall indicate not less than 98 nor more than 102 percent of the water actually passed through the flowmeter within the limits of minimum and maximum flow rates specified in table III. Minimum flow rates shall not be more than 10 percent of the maximum flow rate. Also, there shall not be less than 90 percent registration of the water which actually passes through the flowmeter at a flow rate of 5 percent of the maximum flow rate (see 4.6.2). These registration limits shall be maintained at all water temperatures up to 100°F.

3.7.1.3 Pressure drop. The pressure drop through any class A flowmeter shall not exceed 25 lb/in^2 at the maximum flow rates specified in table III (see 4.6.3).

TABLE III. Class A flowmeter capacities.

Class A flowmeter designation	Maximum flow rate (gallons/minute)	Continuous flow rate (gallons/minute)	Minimum flow rate (gallons/minute)
1	10	5	1
2	30	15	3
3	70	35	7
4	150	75	15
5	500	250	50
6	1000	500	100

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3.7.2 Class B flowmeters.

3.7.2.1 Working pressure. Unless otherwise specified (see 6.2.1), class B flowmeters shall be designed for a minimum working pressure of 100 lb/in^2 (see 4.6.1).

3.7.2.2 Registration. The registration of class B flowmeters shall indicate not less than 97 nor more than 103 percent of the water actually passed through the flowmeter within the limits of minimum and maximum flow rates specified in table IV. Minimum flow rates shall not be more than 20 percent of the maximum flow rate. Also, there shall not be less than 90 percent registration of the water which actually passes through the flowmeter at a flow rate of 10 percent of the maximum flow rate. These registration limits shall be maintained at all water temperatures between 100 and 180°F (see 4.6.2.2).

3.7.2.3 Pressure drop. The pressure drop through any class B flowmeter shall not exceed 20 lb/in^2 at the maximum flow rates specified in table IV (see 4.6.3).

TABLE IV. Class B flowmeter capacities.

Class B flowmeter designation	Maximum flow rate (gallons/minute)	Continuous flow rate (gallons/minute)	Minimum flow rate (gallons/minute)
1	10	5	2
2	20	10	4
3	50	25	10
4	100	50	20
5	250	125	50
6	500	250	100

3.7.3 Class A and B flowmeters.

3.7.3.1 Flow rates. Class A and B flowmeters shall have a maximum flow rate of not less than the maximum flow rate specified in tables III or IV, as applicable. The minimum flow rates shall not be more than the minimum flow rates specified in tables III or IV, as applicable.

3.7.3.2 Continuous flow rate. Class A and B flowmeters shall have a continuous flow rate of not less than 50 percent of the maximum flow rates specified in tables III or IV, as applicable.

3.7.3.3 Intermittent flow. Intermittent flow for class A and B flowmeters shall be considered as operation at the maximum flow rate for not longer than 2 hours out of every 8 hour period.

3.7.4 Shock. The flowmeter shall be resistant to mechanical shock and shall not present any secondary missile hazard (in other words, pieces flying or coming off) when tested as specified in 4.6.4.

3.8 Frost protection. When specified (see 6.2.1), flowmeters shall be furnished with a frost protection device designed to yield under normal freezing conditions to reduce damage to flowmeter parts.

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3.9 Marking.

3.9.1 Identification marking. Flowmeters shall bear the manufacturer's serial number permanently marked on the meter body and on the register box lid. Also, flowmeters shall bear the manufacturer's name, manufacturer's model designation, normal flow range, and the designation "Class A" or "Class B," as applicable. Meter markings shall be in accordance with MIL-STD-130.

3.9.2 Direction-of-flow indication. Flowmeters shall have the direction of flow indicated in raised characters on the flowmeter body.

3.10 Interchangeability. *In no case shall parts be physically interchangeable* or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.11 Technical manuals. When specified by the contract or order (see 6.2.1), technical manuals shall be prepared in accordance with the data ordering document (see 6.2.2).

3.12 Workmanship. The quality of workmanship shall be such as to produce flowmeters that are in accordance with the requirements of this specification and that are so constructed as to ensure proper functioning of all parts of the flowmeter.

3.12.1 Castings and forgings. Castings and forgings shall have no defects or scale and shall not be mismatched. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste, shall not be attempted.

3.12.2 Treatment and painting. Unless otherwise specified (see 6.2.1), the flowmeter shall be treated and painted in accordance with the manufacturer's approved standard practice

3.12.3 Surface finish. Surfaces of castings, forgings, or molded parts, stampings, and welded parts shall be clean and free of sand, dirt, scale, flux, or other extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Terms and definitions. Quality assurance terms and definitions shall be in accordance with MIL-STD-109.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on one flowmeter (see 3.3). The inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory, provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on sample flowmeters selected in accordance with 4.4.2. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.4.1 Inspection lot. Flowmeters of the same class and size, offered to the Government at one time, shall be considered a lot for purposes of inspection.

4.4.2 Sampling for examination. A random sample of flowmeters shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. The acceptable quality level shall be 1.5 percent for major defects and 4.0 percent for minor defects.

4.4.3 Sampling for tests. Sampling is not applicable. Flowmeters passing the examination of 4.5 shall be tested in accordance with 4.6.

4.5 Examination. Flowmeters selected in accordance with 4.4.2 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted as specified in table V. Defective flowmeters shall be rejected and, if the number of defective flowmeters in any sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

TABLE V. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Class or type not as specified.
102	Missing, broken, or improper parts.
103	Flowmeters not rated in accordance with system requirements.
104	Materials not as specified.
105	End connections not as specified.
106	Maximum indication unit of initial register circle or wheel and respective minimum capacity or register nonconforming.

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TABLE V. Classification of defects. - Continued

Categories	Defects
Major	
107	Horizontal type registers not furnished with covers; do not protect delicate and fragile parts.
108	Dial face difficult to read, cloudy, fogs easily, not adequately protected.
109	Loose, unbalanced, or improperly meshing gears.
110	Defective castings or forgings.
111	Marking not as specified.
113	Wearing parts not accessible for service.
114	Strainers furnished do not fit; size of strainer opening incorrect.
115	Check valve furnished to prevent flowmeter damage during flow reversal not furnished or not functioning properly.
Minor	
201	Surface finish not as specified.
202	Painted surfaces scratched or uncovered.

4.6 Method of inspection.

4.6.1 Hydrostatic leakage. Sample flowmeters selected in accordance with 4.4.3 shall be subjected to an internal hydrostatic pressure for 30 minutes equal to 150 percent of the working pressure specified in either 3.7.1.1 or 3.7.2.1, as applicable.

4.6.2 Calibration. Sample flowmeters selected in accordance with 4.4.3 shall be subjected to a calibration as specified in 4.6.2.1 and 4.6.2.2.

4.6.2.1 Test conditions. Class A flowmeters shall be tested at a water temperature of 95 ± 5°F at 5, 10, 15, 25, 50, 75, and 100 percent of the maximum flow rate specified in table III. Class B flowmeters shall be tested at water temperatures of 105 ± 5°F and 175 ± 5°F at 10, 15, 20, 25, 50, 75 and 100 percent of the maximum flow rate specified in table IV.

4.6.2.2 Registration. Registration shall be verified in accordance with 3.7.1.2 or 3.7.2.2, as applicable, at the test conditions specified in 4.6.2.1.

4.6.3 Pressure drop. Sample flowmeters selected in accordance with 4.4.3 shall be tested to verify conformance to 3.7.1.3 or 3.7.2.3, as applicable.

4.6.4 Shock. The flowmeter shall be subjected to grade B, class II, lightweight classification, type B shock specified in MIL-S-901.

4.7 Test failures. Test failures are defined as follows:

(a) Hydrostatic leakage test: When tested as specified in 4.6.1, leakage indicates that the working pressure is less than specified.

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(b) Calibration: When tested as specified in 4.6.2, the accuracy of registration at the designated flow rates is not within specified limits.
(c) Pressure drop: When tested as specified in 4.6.3, the pressure drop exceeds the maximum specified.
(d) Shock test: Components or pieces coming loose or becoming noticeably loosened giving indication that this item may constitute a missile hazard to personnel and other equipment.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. Openings in the flowmeters shall be sealed with plugs or blank flanges, as applicable. The flowmeters shall be packaged in accordance with method III of MIL-P-116. Delicate or fragile parts shall be wrapped with cellulosic cushioning material conforming to MIL-STD-1186. Flowmeters weighing 70 pounds or less shall be individually packaged in fiber boxes conforming to PPP-B-636. Box closures shall be as specified in the appendix to the box specification. Flowmeters exceeding 70 pounds shall be individually placed in containers for the level of packing specified (see 6.2.1).

5.1.2 Level C. Flowmeters shall be preserved and packaged in accordance with the manufacturer's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Flowmeters, packaged as specified (see 6.2.1), shall be packed in snug fitting wood-cleated fiberboard, wood-cleated plywood, nailed wood, or fiberboard boxes conforming to PPP-B-591 (class II), PPP-B-601 (overseas type), PPP-B-621 (class 2) of PPP-B-636, respectively, at the option of the contractor. Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for fiber boxes conforming to PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 1-1/2-inch wide tape conforming to A-A-1683. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Wood or wood-cleated boxes exceeding 200 pounds shall be modified by the addition of nominal 2 by 4 inch wood skids in accordance with PPP-B-621.

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5.2.2 Level B. Flowmeters, packaged as specified (see 6.2.1), shall be packed in snug-fitting, wood-cleated fiberboard, wood-cleated plywood, nailed wood, or fiberboard boxes conforming to PPP-B-591 (class I), PPP-B-601 (domestic type), PPP-B-621 (class 1), or PPP-B-636 (domestic class), respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Wood or wood-cleated boxes exceeding 200 pounds shall be modified by the addition of nominal 2 by 4 inch wood skids in accordance with PPP-B-621.

5.2.3 Level C. Flowmeters shall be packed to ensure carrier acceptance and safe delivery to destination. Containers shall comply with the Uniform Freight Classification rules or other carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2.1), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The flowmeters covered by this specification are intended for operation in freshwater systems at temperatures up to 180°F and for flow rates up to 1000 gallons per minute.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size of flowmeter (see 1.2 and tables II, III, and IV).
(c) Resettable counter, when required (see 3.1).
(d) First article when required (see 3.3).
(e) When measuring chamber is not to be manufacturers standard material (see 3.6.3).
(f) Type of register, if other than specified (see 3.6.5).
(g) Type of strainer, if other than specified (see 3.6.6).
(h) Type end connections, if other than specified (see 3.6.8).
(i) Working pressure, if other than specified (see 3.7.1.1 or 3.7.2.1).
(j) Frost protection, when required (see 3.8).
(k) Technical manuals, when required (see 3.11).
(l) Treatment and painting, if other than specified (see 3.12.2).
(m) Levels of preservation, packaging, packing, and marking (see 5.1, 5.2, and 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified

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below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.11	Manuals, technical, standard basic issue	DI-M-2044	Type I of MIL-M-15071

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Definitions.

6.5.1 Approved standard commercial product. The manufacturer's approved standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs or brochures and represents the latest approved production models. The principal contracting officer will review requirement 3.2 for compliance with the provision of part II of the Federal Acquisition Regulation.

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6.5.2 Registration. Registration is the ratio of the volume of water indicated on the register to the volume of water which actually passes through the flowmeter at a specified flow rate.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6680-N204)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

SHIPBOARD USE

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-2082C(SH)	LIQUID, COLD WATER AND HOT WATERT YPE, NAVAL METERS, FLOW, MECHANICAL, VOLUMETRIC POSITIVE DISPLACEMENT,

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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